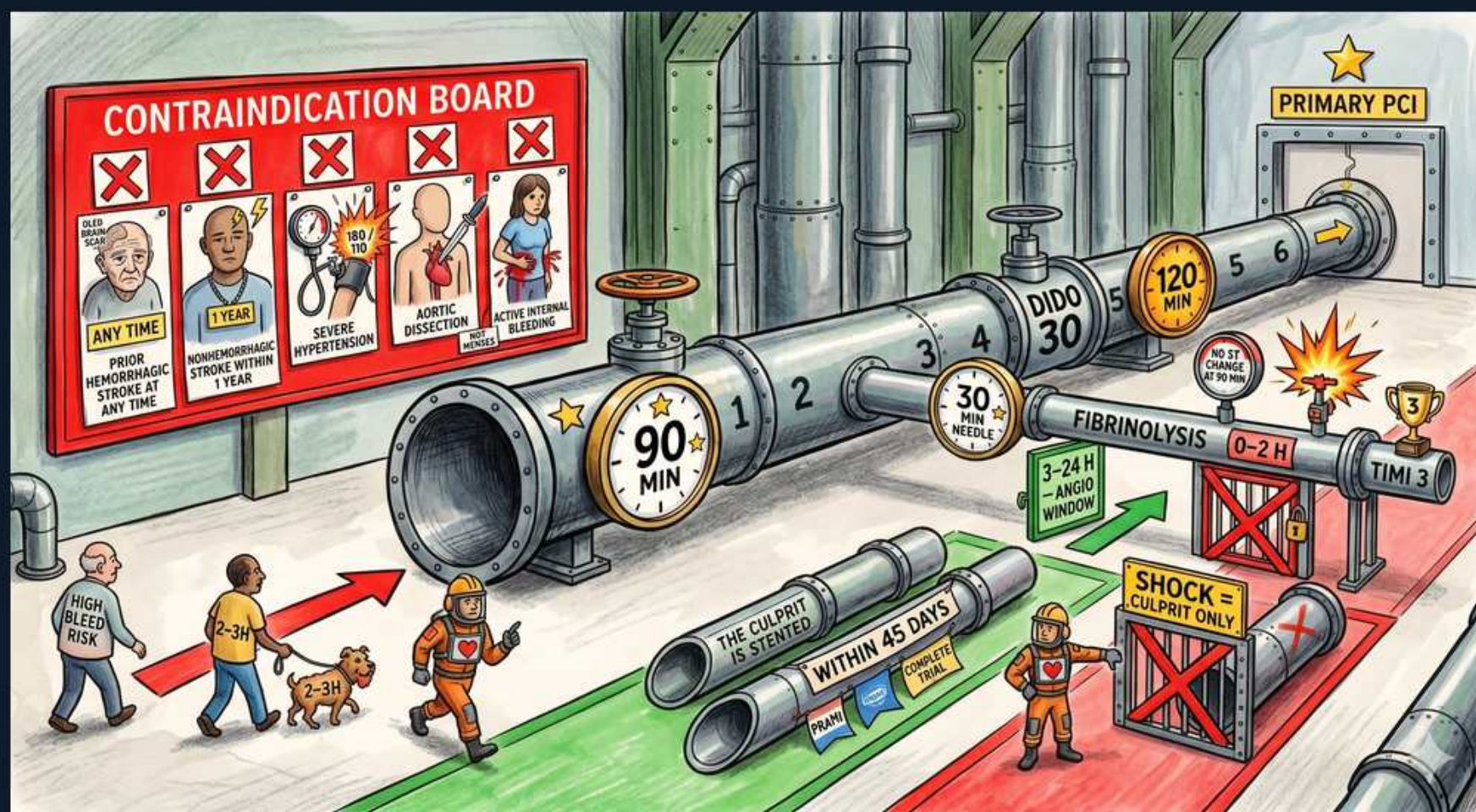


STEMI — Reperfusion Timing, PCI vs Lysis & Contraindications



1. Contraindication board

WHAT YOU SEE

'CONTRAINDICATION BOARD' billboard with X panels

WHAT IT ENCODES

Absolute contraindications to fibrinolysis: any prior intracranial hemorrhage, ischemic stroke <3 months, known structural cerebral vascular lesion/malignancy, suspected aortic dissection, active bleeding, significant closed-head trauma <3 months.

BOARD TRAP

These are contraindications to LYSIS, not to PCI — a patient who can't get lytics should still go for primary PCI.

2. Prior hemorrhagic stroke — any time

WHAT YOU SEE

Panel 'PRIOR HEMORRHAGIC STROKE AT ANY TIME'

WHAT IT ENCODES

ANY prior intracranial hemorrhage is an absolute, lifetime contraindication to thrombolysis. Refer for primary PCI instead.

BOARD TRAP

Treating a remote hemorrhagic stroke as 'old, so safe' — prior ICH contraindicates lytics permanently.

3. Ischemic stroke within 1 year

WHAT YOU SEE

Panel 'ISCHEMIC STROKE WITHIN YEAR'

WHAT IT ENCODES

Ischemic stroke within 3 months is an absolute contraindication; the broader caution extends toward recent strokes. Older ischemic stroke (>3 mo) is a relative contraindication.

BOARD TRAP

The hard cutoff for absolute contraindication is ischemic stroke within 3 months — beyond that it becomes relative.

4. Severe hypertension

WHAT YOU SEE

Panel 'SEVERE HYPERTENSION'

WHAT IT ENCODES

Severe uncontrolled HTN (SBP >180 / DBP >110) is a relative contraindication to lysis — control BP first or choose PCI. Reflects intracranial bleed risk.

BOARD TRAP

Severe HTN is a RELATIVE contraindication — it can be controlled, unlike the absolute ones; PCI is preferred.

5. Aortic dissection

WHAT YOU SEE

Panel 'AORTIC DISSECTION'

WHAT IT ENCODES

Suspected aortic dissection is an absolute contraindication — lytics are catastrophic. Dissection can mimic/cause STEMI (RCA ostium); always exclude before reperfusion if suspected.

BOARD TRAP

Lysing a 'STEMI' that's actually aortic dissection extending to the coronary ostium — fatal; image first if suspected.

6. Active internal bleeding

WHAT YOU SEE

Panel 'ACTIVE INTERNAL BLEEDING'

WHAT IT ENCODES

Active internal bleeding (excluding menses) is an absolute contraindication. Recent major surgery/trauma and bleeding diathesis are relative/absolute depending on severity.

BOARD TRAP

Giving fibrinolytics to a patient with a recent GI bleed — active bleeding is an absolute contraindication.

7. Primary PCI — gold standard

WHAT YOU SEE

'PRIMARY PCI' gold sign + trophy

WHAT IT ENCODES

Primary PCI is the preferred reperfusion strategy when achievable within timing goals — superior to lysis in mortality, reinfarction, and stroke. Always preferred in shock or contraindication to lytics.

BOARD TRAP

Choosing lysis over available timely PCI — primary PCI is superior; lysis is the fallback when PCI isn't reachable in time.

8. 90 min — door-to-balloon

WHAT YOU SEE

Gauge '90 MIN'

WHAT IT ENCODES

Door-to-balloon ≤90 minutes is the target when the patient presents to a PCI-capable hospital. Faster reperfusion = more myocardium salvaged ('time is muscle').

BOARD TRAP

Tolerating delays beyond 90 min at a PCI center — door-to-balloon goal is ≤90 minutes.

9. DIDO 30 — transfer goal

WHAT YOU SEE

Gauge '30 / DIDO'

WHAT IT ENCODES

Door-in-door-out (DIDO) ≤ 30 min at a non-PCI hospital — minimize time before transfer. Total first-medical-contact-to-PCI should be ≤ 120 min for transfer to beat lysis.

BOARD TRAP

Sitting on a STEMI at a non-PCI center — DIDO should be ≤ 30 min to keep total FMC-to-device ≤ 120 min.

10. 120 min — transfer cutoff

WHAT YOU SEE

Gauge '120 MIN'

WHAT IT ENCODES

If PCI can be delivered within 120 min of first medical contact, transfer for PCI; if not, give fibrinolysis within 30 min and transfer (pharmacoinvasive strategy).

BOARD TRAP

Transferring for PCI when it will take > 120 min — give lytics first, then transfer (don't lose myocardium waiting).

11. 30-min needle — door-to-needle

WHAT YOU SEE

'30 NEEDLE'

WHAT IT ENCODES

If fibrinolysis is chosen, door-to-needle ≤ 30 minutes. Greatest benefit within the first 0–3 hours of symptom onset; refer for rescue/early PCI afterward.

BOARD TRAP

Delaying lytic administration — door-to-needle target is ≤ 30 minutes when PCI is not timely.

12. Fibrinolysis 0–2h golden window

WHAT YOU SEE

'FIBRINOLYSIS 0–2 H — GOLDEN WINDOW; if it can change outcome'

WHAT IT ENCODES

Lysis is most effective within 0–2(–3) h of symptom onset (greatest clot dissolution, abortable infarct). Beyond 12 h, lysis is not indicated. After lysis, routine early angiography (3–24 h).

BOARD TRAP

Giving lytics > 12 h after symptom onset — past the window there's no benefit; pursue PCI based on ongoing ischemia.

13. TIMI 3 — successful reperfusion

WHAT YOU SEE

'TIMI 3' trophy

WHAT IT ENCODES

TIMI 3 flow = full, normal coronary perfusion, the goal of reperfusion. Failed lysis (persistent ST elevation, ongoing pain) → rescue PCI.

BOARD TRAP

Assuming lysis worked without checking reperfusion criteria — failed reperfusion (no ST resolution) mandates rescue PCI.

14. Within 45 days — culprit only

WHAT YOU SEE

'WITHIN 45 DAYS / THE CULPRIT IS STENTED' pipe

WHAT IT ENCODES

In primary PCI for STEMI, the culprit lesion is treated acutely; complete (non-culprit) revascularization is increasingly favored but typically staged. In shock, treat the culprit only.

BOARD TRAP

Stenting all lesions during cardiogenic shock — CULPRIT-SHOCK showed culprit-only PCI improves outcomes in shock.

15. Shock = culprit only

WHAT YOU SEE

'SHOCK = CULPRIT ONLY' red X crate

WHAT IT ENCODES

In cardiogenic shock complicating STEMI, perform culprit-lesion-only PCI; multivessel intervention in the same sitting increased death/renal failure (CULPRIT-SHOCK trial).

BOARD TRAP

Doing complete multivessel PCI in a STEMI patient in shock — culprit-only is the evidence-based approach.

16. High bleed risk — favor PCI

WHAT YOU SEE

'HIGH BLEED RISK' man + dog

WHAT IT ENCODES

Patients at high bleeding risk (recent surgery, anticoagulation, advanced age, bleeding diathesis) should preferentially receive primary PCI rather than fibrinolysis to avoid hemorrhagic complications.

BOARD TRAP

Choosing lysis in a high-bleeding-risk patient — primary PCI avoids the systemic hemorrhage risk of fibrinolytics.